

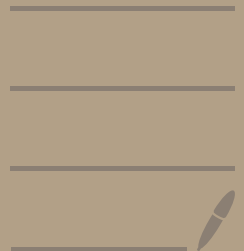
Lec 13 , Autonomous system.

phase plane.

Linear stability

Ref : Sec 9.1, 9.2, 9.3. [BDM] (Elementary case)

(for mid-term).



I: Basic concept.

Consider 1st order linear system.

$$\frac{d}{dt} \vec{x} = \vec{F}(t, \vec{x})$$

Autonomous: If $\vec{F}$ is independent of t

E.g.

$$\frac{d}{dt} \vec{x} = A\vec{x}$$

$$\frac{dx}{dt} = px + qx^2, \quad p, q \text{ constant.}$$

Consider

$$\frac{d}{dt} \vec{x} = \vec{F}(\vec{x}) \quad (1)$$

Equilibrium - x^* is equilibrium to (1) if $\vec{F}(x^*) = 0$.

Also called critical points.

Let $x(t)$ be a solution to (1). we call $(t, x(t))$ a trajectory

For $n \geq 2$

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{pmatrix}$$

The x_1, x_2 -plane is called phase plane.

Representative set of trajectories is called phase portrait

$$\frac{\bar{G}-f,}{\cdot} \quad \left\{ \begin{array}{l} \frac{d}{dt} x = x^2 - 2y \\ \frac{d}{dt} y = x + y \end{array} \right.$$

Find equilibrium

$$\frac{d}{dt} x = x^2 - b^2.$$

• [A]. Phase plane of 2×2 linear system.

$$\frac{d}{dt} \vec{x} = A \vec{x} \quad \vec{x} \in \mathbb{R}^{2 \times 1}, \quad A \in \mathbb{R}^{2 \times 2}.$$

• 3 cases

1. 2 real roots (not equal)

2. complex root

3. repeated real roots.

1. $x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$

discuss sign of λ_1, λ_2 .

Sink
Source
Saddle.

2. $x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2 \quad \lambda = \alpha \pm i\beta$

$$= e^{\alpha t} \left[(\cos(\beta t) \vec{a} - \sin(\beta t) \vec{b}) + i(\vec{b} \sin(\beta t) + \vec{a} \cos(\beta t)) \right]$$

$$e^{\alpha t} (\cos(\beta t) + i \sin(\beta t)) (\vec{a} + i \vec{b})$$

when $\alpha = 0$. Periodic

3. v_1, v_2 linearly independent

$$x(t) = e^{\lambda t} (c_1 v_1 + c_2 v_2) \rightarrow \text{straight line}$$

only one eigenvector

$$x(t) = e^{\lambda t} (c_1 v_1 + c_2 v_2)$$



(not easy to draw)

Refer Pics in Sec 9.1 (BDM)

IV Stability: Summarize the Stability.

$$(2) \quad \frac{d}{dt} \vec{X} = \vec{F}(\vec{X})$$

Let X^* be a critical point: $\vec{F}(X^*) = 0$

Stable: If $\forall \epsilon > 0, \exists \delta > 0$ s.t

$\forall$ solution $X = \phi(t) \rightarrow 0$ (2), if

$$|\phi(0) - X^*| < \delta,$$

$$\Rightarrow |\phi(t) - X^*| < \epsilon, \quad \forall t \geq 0.$$

Unstable: If X^* is not stable.

Asymptotic Stable if stable + $\exists \delta_0$ s.t.

$$|\phi(0) - X^*| < \delta_0$$

then

$$\lim_{t \rightarrow \infty} \phi(t) = X^*.$$

Linear system.

Consider

$$\frac{d}{dt} X = AX, \quad A \in \mathbb{R}^{n \times n}.$$

Thm 1: (Generalize Thm 9.3.1).

Consider $X^* = 0$. $\lambda_1, \dots, \lambda_n$ eigenvalues of A .

(1) If $\operatorname{Re}(\lambda_i) < 0, \forall i \Leftrightarrow X^*$ is asymptotic stable

(2) $\exists i$ s.t. $\operatorname{Re}(\lambda_i) > 0 \Rightarrow$ Unstable

(3) If $\operatorname{Re}(\lambda_i) \leq 0, \forall i \Leftrightarrow X^*$ stable.

(4) $\exists i$ s.t. $\operatorname{Re}(\lambda_i) = 0 \Rightarrow$ Not asymptotic stable.

pf: Use Jordan form. See class.

• Thm 2. $A \in \mathbb{R}^{n \times n}$. If $\operatorname{Re}(\lambda_i) \leq -\lambda < 0$.
Then we have

$$X(t) = e^{At} X_0, \quad \max_i |X_i(t)| \leq C \cdot e^{-\lambda t} \max_i |X_{0,i}|.$$

Pf:

Step 1: Use Jordan normal form, $A = U \cdot J \cdot U^{-1}$

$$e^{At} X_0 = U \begin{pmatrix} e^{J_{11}t} & & \\ & \ddots & \\ & & e^{J_{nn}t} \end{pmatrix} U^{-1} \cdot X_0$$

Step 2: Each block J_i .

$$|(e^{J_{ii}t})_{ij}| \leq C \cdot e^{-\lambda t} \cdot t^n.$$

• Constant C depends on A
• can change from line to line.

$$\Rightarrow |(e^{Jt})_{ij}| \leq C \cdot e^{-\lambda t} t^n$$

Step 3:

$$\begin{aligned} |[U \cdot e^{Jt} U^{-1}]_{ij}| &\leq n^3 \cdot \max |U_{ij}| \cdot C \cdot e^{-\lambda t} t^n \\ &\leq C \cdot e^{-\lambda t} t^n. \end{aligned}$$

$$\Rightarrow |(e^{At})_{ij}| \leq C \cdot e^{-\lambda t} t^n.$$

$$\Rightarrow \max |X_i(t)| = \max |(e^{At} X_0)_i| \leq C \cdot e^{-\lambda t} t^n \cdot \max |X_{0,i}|$$

Ex: $e^{-\lambda t} t^n \leq C_n \cdot e^{-\alpha t}, \quad \forall \alpha > \lambda.$

Example :

$$A(t) = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 3 & \\ & & -2 \end{pmatrix}$$